The Fabric Filter

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ABSTRACT

Air pollution is a critical challenge impacting human health, ecosystems, and climate, particularly in regions like Egypt, where pollution levels frequently exceed safe limits. This project addresses this issue by leveraging locally-sourced waste materials to develop a two-stage air purification system. The system is designed to mitigate fine particulate matter (PM2.5) and carbon monoxide (CO), pollutants commonly found in urban environments, and cigarette smoke. To align with the capstone challenge requirements, the filtration materials include reprocessed polyester derived from textile waste and baking soda, which acts as a corrosion inhibitor and adsorbent for CO. During testing, the system demonstrated a 51.7% reduction in pollutant concentration within 7 minutes, far surpassing the minimum requirement of a 20% reduction within 10 minutes. Pollutant levels were monitored using calibrated sensors before and after filtration to validate system performance. The prototype is scalable, durable, and designed to handle higher pollution levels, meeting the capstone's emphasis on practical and impactful solutions. This project not only showcases the feasibility of converting waste into valuable products for air purification but also contributes to Egypt's broader efforts to combat air pollution and its associated public health challenges. A video demonstration and live prototype performance will further illustrate the system's effectiveness in real-time scenarios, ensuring compliance with all project deliverables and evaluation criteria.

INTRODUCTION

Egypt faces significant environmental challenges, with air pollution increasing by 20% over the past decade. This increase in pollution levels is primarily due to the presence of harmful pollutants such as nitrogen dioxide, carbon monoxide, sulfur dioxide, and particulate matter. These pollutants pose serious health risks, including respiratory disorders, lung cancer, and premature death. Moreover, the impact of air pollution extends beyond human health, contributing to accelerating climate change through ozone depletion and intensifying global warming. Addressing these issues is critical to protecting public health, ensuring safety, and promoting sustainable environments. Previous air purification methods, such as electro-spun nanofiber polyvinyl alcohol/cellulose (PVA/CNCs) filters, have demonstrated high filtration efficiency and recyclability potential. However, they also have significant limitations. These methods often involve complex manufacturing processes that require specialized equipment, making them expensive and difficult to scale up. Additionally, they are sensitive to environmental conditions such as high humidity, which can reduce their effectiveness and practicality in real-world applications. High production costs and the need for precision manufacturing limit the scalability of these approaches, making them unsuitable for large-scale application in areas such as Egypt, where affordable and accessible solutions are critical. To address these challenges, the proposed solution offers a more practical and sustainable approach to air purification. The goal is to reduce air pollution by at least 20% within 10 minutes for air samples ranging from 600 to 1500 ml. This design leverages locally available waste materials to create a durable, flexible, and environmentally friendly filtration system that does not rely on pre-manufactured filters or hazardous materials. The filtration system consists of materials such as polyester fabric, wire mesh, baking soda, and coffee grounds. These materials work together to filter out particulate matter, neutralize harmful chemicals, and eliminate odors. The use of locally sourced waste materials helps reduce production costs and increase the sustainability of the system, making it an ideal solution to Egypt's air pollution crisis. The system's design ensures efficiency, practicality, and scalability, making it suitable for large-scale use in a variety of environments, especially in heavily polluted urban areas. By avoiding the need for expensive, specialized equipment and complex manufacturing processes, this solution is cost-effective and easy to implement. The innovative use of readily available materials also makes the system adaptable to different environmental conditions, ensuring that it can be used effectively in a variety of environments. Overall, this approach not only meets the design requirements, but also provides a sustainable, affordable, and scalable solution to address the urgent air quality challenges facing Egypt today.

MATERIALS

Table (1): Materials

Materials	The Cost	picture
Arduino UNO board	460 L.E	
MQ135 sensor	90 L.E	
Polyester fabric	15 L.E	
Coffee	20 L.E	
Baking soda	10 L.E	
Plastic box	25 L.E	
2.5L bottle	10 L.E	
Hose	30 L.E	
Mesh	15 L.E	

Total cost: 675 L.E

METHODS

The prototype was built in several steps. Step 1: Prototype devices were built using mesh and polyester filters that were glued and fixed with adhesive. Coffee grounds and baking soda were chosen as the primary air purifiers due to their effectiveness in reducing air pollutants. A plastic container served as the main body of the prototype, with a cut hole to connect a hose to draw air. A separate bottle was used to store cigarette smoke (carbon monoxide gas), which facilitated controlled experimental analysis. The bottle was connected to the hose to direct polluted air into the prototype.

Step 2: To measure air quality, an MQ-135 sensor was integrated with an Arduino UNO board. To ensure accurate measurements, we connected the sensor and left it running for three to five minutes before starting the test plan.

Step 3: In the final stage, the prototype was tested in a real-world simulation. The contaminated air was pumped from the 2500 ml bottle into the tank through the connected hose. The purification process occurred as the air passed through the filter material inside the tank. To evaluate the performance of the prototype, sensor readings were taken before and after filtration. The results were compared to assess the reduction in airborne pollutants and verify the effectiveness of the design.

Fig (1): Air container

Test Plan:

The test began by verifying the sensor's functionality by measuring the natural air quality, which was recorded at 297 PPM. The sensor was then placed inside the bottle containing carbon monoxide gas from burning cigarettes, where it measured a concentration of 907 PPM.

Fig (2): Filter

The contaminated air was then pushed through the hose into the main 2500 ml container. Over seven minutes, the air purifier filtered the carbon monoxide gas concentration from 907 PPM to 438 PPM, achieving a reduction of 469 PPM. This corresponds to a filtration efficiency of 51.7%.

These results demonstrate the effectiveness of the prototype in meeting the design requirements and its ability to significantly reduce airborne pollutants, confirming the success of the project.

Fig (3): Arduino board

RESULTS

Following the construction of the test plan, the readings and outputs of the project were carefully collected and saved in a database in the format of EXCEL Sheets, the data was analyzed and been compared with the design requirements to ensure that the project meets the design requirements successfully.

Table (2): Data calculations of the test plan

Air sample before filtration	907 PPM
Air sample after filtration	438 PPM
Filtered air concentration	469 PPM
Concentration/minute	67 PPM/min
Time	7 min

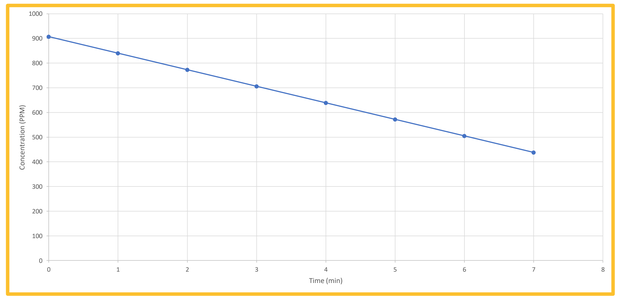


Fig (4): Concentration change

Table (3): Concentration change

Time (min)	0	1	2	3	4	5	6	7
Concentration (PPM)	907	840	773	706	639	572	505	438

ANALYSIS

Based upon the explosive population growth, car exhausts, industrial fumes and uncontrolled deforestation; a significant climate change and rigorous air pollution have taken place. From that relation, these actions are directly proportional to the air pollution and climate change globally. In such a way, these factors directly contribute to global warming.

Polyester fabric filters present an optimal solution for this substantial challenge; in the purist of mitigating and controlling reasons and effects of air pollution, climate change and global warming. In particular for its low cost, ease of usage, sustainability, high recycling ability and ease of maintenance; first and foremost, its high efficiency of filtration. Such that the efficiency of the mentioned filter as illustrated in figure (5) was about 51.7% in 7 minutes for a sample of 2500 ml, with a rate of Co gas purifying of 67 PPM/minute as shown in figure (6). In spite of the positive results that have been recorded, for instance high efficiency in contaminants purification and its sustainability due to the waste materials used in the prototype; negative results have been also faced for example, sensor overheat as it had been working for about 5 minutes before the test plan to ensure high precision and 7 minutes of the test plan, with a total time of 12 minutes. After the test plan results were analyzed, it was concluded that the project successfully met the design requirements of air purified percentage (20% or above) and time of purifying (10 minutes or less). Additionally, the sustainability requirements were met by using more than one waste material.

$$efficiency = \frac{output}{input} \times 100$$

Fig(5): Efficiency equation

$$Rate\ of\ purification = \frac{PPM}{time}$$

Fig(6): Rate of concentration equation

Following the construction of the prototype, a test plan was developed to be compared with the design requirements. The initial area of inquiry was on the PPM of the polluted air sample as shown in figure (7) and the PPM of it after filtration as illustrated in figure (8), which were 907 PPM before and 438 PPM after the purification process. Moreover, the amount of PPM purified per minute which was 67 PPM/minute according to figure (6), efficiency which was 51.7% based on figure (5) and time of purifying which was 7 minutes.

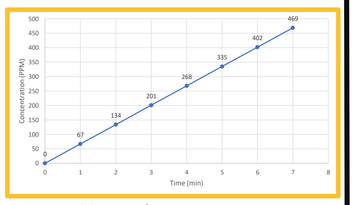
1	Concentration (PPM)
2	898
3	901
4	904
5	905
6	906
7	907
8	907
9	907
10	907

Fig(7): Readings before filtration

1	Concentration (PPM)
2	443
3	443
4	442
5	441
6	440
7	439
8	439
9	439
10	438

Fig(8): Readings after filtration

As reflected in figure (9), the rate of air purification is represented in a linear function with time (t) on the x-axis and concentration of purified particles (PPM), which is 67 PPM/minute. The total concentration of purified air can be calculated using the equation $\frac{PPM}{t} \times total\ time$, which equals $67 \times 7 = 469$ PPM.



Fig(9): Rate of concentration per minute

From all the previous, the project had succeeded to prove that it is a good solution for the grand challenge facing the world, which is improving the environmental health and well-being and decrease the air pollution, climate change and global warming.

Learning Outcomes:

Table (4): Learning outcomes table

Learning Outcome	Description
CH 2.01	Calculating the concentration and understanding the filtration process of the particles per million (PPM)
CH 2.04	The rate of reaction of filtration depends on the concentration of the reactants to enhance the filtration process speed
MA 2.01	Using functions to translate the sensor's readings into graphs
ES 2.01	Temperature increases the particles bounding speed, causing an increase in the filtration efficiency of the small particles
CS 2.02	Constructing a database to analyze and save the concentration outputs

The highlighted text with this color is important!

CONCULSION

Following testing, the prototype was subjected to the MQ-135 air quality sensor, generating astonishing results on its effectiveness in purifying air impurities from pollutants. The prototype was able to reduce the pollutants concentration that attained a pollution concentration of about 907 PPM reduced down to about 438 PPM during 7 minutes, hence giving purification efficiency of 51.7%. The reduction indicates the efficiency of our design that has layers of polyester fabric with baking soda and coffee for filtration and chemical cleansing. The volume of the air sample was at 2500 ml, as stated in the design specifications. It remains, therefore, ecologically friendly while using waste materials and incurring none of the impacts on the environment. Our solution is much simpler, cheaper, and scalable than all other methods of purification of air. Our prototype is designed according to those design criteria and does match the aims of the Grand Challenge in Egypt on air contamination, utilizing locally sourced waste materials for doing good to the environment. All in all, the project has completed its goal, and such is the auspicious avenue to reduce air pollution, with further enhancement possibilities.

Recommendation

To enhance the performance and user experience of the project, it is suggested to implement the following improvements:

Multi-Stage Filtration Mechanism: Adopt a modular multi-layer filtration system. For instance, combine pre-filters for larger particles, HEPA filters for fine particles, and activated carbon for chemical pollutants. Such a system will effectively handle diverse pollutants and improve indoor air quality comprehensively.

Public Engagement and Awareness: Incorporate public education initiatives by developing interactive displays, conducting workshops, and distributing easy-to-read pamphlets about air pollution and sustainable practices. This will not only improve awareness but also promote adoption of air purification solutions.

Advanced IoT Integration: Develop an Internet of Things (IoT)-enabled platform that integrates multiple sensors to monitor air quality parameters, such as PM2.5, CO, and VOCs. This system can provide real-time updates via a user-friendly mobile app or web interface, ensuring accessibility and control from remote locations.

Aerodynamic Optimization: Use advanced simulation techniques, such as Computational Fluid Dynamics (CFD), to design an airflow system that ensures even pollutant distribution across the filter media. This will maximize the contact surface area between pollutants and activated carbon for improved filtration efficiency.

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